

3. Scope of Testing

Overview

The scope of testing defines the boundaries and coverage of the API testing activities performed in this project. This project focuses on validating the functionality, reliability, and performance of the Fake Store REST API using Postman and Newman CLI.

The testing scope includes verification of various API endpoints related to Products, Categories, Carts, Users, and Authentication modules. The objective is to ensure that each API endpoint behaves as expected, returns accurate data, and supports the required business functionality of an e-commerce application.

The testing activities are limited to API-level testing and do not include frontend UI testing, database testing, security testing, or performance/load testing.

Modules Covered in Testing

1. Authentication Module

The Authentication module is tested to verify user login functionality and token generation.

Testing Coverage:

- Verify successful login with valid credentials.
- Verify API response status code.
- Verify token generation after successful login.
- Verify response body structure.
- Validate response time.

API Covered:

- Login User
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2. Products Module

The Products module is tested to ensure that product-related operations function correctly.

Testing Coverage:

- Retrieve all available products.
- Retrieve a specific product using Product ID.
- Add a new product.
- Update an existing product using PUT method.

- Partially update product details using PATCH method.
- Delete an existing product.
- Validate response data and status codes.
- Verify response times.

APIs Covered:

- Get All Products
 - Get Single Product
 - Add New Product
 - Update Product (PUT)
 - Update Product (PATCH)
 - Delete Product
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3. Categories Module

The Categories module is tested to validate category-related operations.

Testing Coverage:

- Retrieve all available product categories.
- Retrieve products belonging to a specific category.
- Validate category data returned by the API.
- Verify response structure and status codes.

APIs Covered:

- Get All Categories
 - Get Products in Category
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4. Cart Module

The Cart module is tested to verify shopping cart functionality.

Testing Coverage:

- Retrieve all cart details.
- Retrieve a specific cart.
- Add products to cart.
- Update cart information.
- Partially update cart details.
- Delete cart records.
- Validate response structure and status codes.

APIs Covered:

- Get All Carts
 - Get Single Cart
 - Add to Cart
 - Update Cart (PUT)
 - Update Cart (PATCH)
 - Delete Cart
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5. Users Module

The Users module is tested to validate user information retrieval.

Testing Coverage:

- Retrieve all users.
- Retrieve a specific user using User ID.
- Validate user details returned in response.
- Verify response structure and response status.

APIs Covered:

- Get All Users
 - Get Single User
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Validation Areas Covered

The following validations are performed during API testing:

Status Code Validation

Verification that APIs return the expected HTTP status codes.

Examples:

- 200 OK
 - 201 Created
 - 400 Bad Request
 - 404 Not Found
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Response Body Validation

Verification that the response contains correct and expected data.

Examples:

- Product Name
 - Product Price
 - Category Name
 - User Details
 - Cart Information
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JSON Structure Validation

Verification that API responses follow the expected JSON format and contain required fields.

Examples:

- id
 - title
 - price
 - category
 - description
 - image
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Response Time Validation

Verification that APIs respond within acceptable performance limits.

Example:

- Response time should be less than 2000 milliseconds.
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Environment Variable Validation

Verification that environment variables are properly stored and utilized.

Examples:

- Base URL
 - Product ID
 - User ID
 - Cart ID
 - Authentication Token
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Automated Test Script Validation

Verification of JavaScript assertions implemented in Postman.

Examples:

- Status code assertions
 - Response body assertions
 - Response time assertions
 - Environment variable assertions
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Testing Types Included

The following testing types are included within the project scope:

Functional Testing

Validates whether API endpoints perform the expected business operations correctly.

CRUD Testing

Validates Create, Read, Update, and Delete operations across multiple modules.

Integration Testing

Validates communication between API requests and server responses.

Regression Testing

Ensures existing functionality continues to work after updates or modifications.

Smoke Testing

Performs basic verification to confirm that major API functionalities are working correctly.

Activities Included in Scope

The following activities are part of this project:

- API Request Creation
- API Execution
- Status Code Verification
- Response Validation
- JSON Validation
- Collection Creation
- Environment Setup

- Collection Runner Execution
 - Newman CLI Execution
 - HTML Report Generation
 - Test Result Analysis
 - Defect Identification
 - Documentation Preparation
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Activities Excluded from Scope

The following testing activities are not covered in this project:

- UI Testing
 - Database Testing
 - Security Testing
 - Penetration Testing
 - Load Testing
 - Stress Testing
 - Accessibility Testing
 - Mobile Application Testing
 - Cross-Browser Testing
 - Automation Framework Development using Selenium
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Scope Summary

The scope of this project is focused on end-to-end API testing of the Fake Store REST API using Postman and Newman CLI. The testing covers Authentication, Products, Categories, Carts, and Users modules, including CRUD operations, response validation, status code verification, response time checks, automated test execution, and report generation. The project demonstrates practical API testing techniques commonly used in real-world software testing environments.